

Elastic ML Inference Serving with Custom Autoscaling

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Goal

- Implement a custom autoscaler for an ML inference service
- Kubernetes + Docker + Prometheus + Redis
- Compare against Kubernetes HPA (70% and 90%)]
- Achieve server-side latency of $< 0.5s$ for the queries

System Architecture

Diagram

Load Tester --> Dispatcher (Flask + Redis) --> ML Inference Replicas
(ResNet18)

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Prometheus <----- Metrics (port 8000, 8001)

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Custom Autoscaler (reads Prometheus, scales via kubectl)

System Architecture

Flow

- Load tester generates requests:
- Dispatcher queues requests
- Replicas process images
- Prometheus collects metrics
- Autoscaler changes replica count every 15 seconds

Metrics Used

P99 Latency	Inference execution time inside worker pods
Queue Size	Waiting requests inside dispatcher
Replica Count	Number of active ML pods

Note: Queue size was the main indicator of overload because P99 latency measured only inference execution time.

Implementation of Custom HPA

Custom Autoscaler Logic

```
# === Scaling Logic ===
def compute_target_replicas(p99_latency, queue_size, current_replicas):
    if queue_size is None:
        queue_size = 0

    if queue_size == 0:
        return max(current_replicas - 1, MIN_REPLICAS)

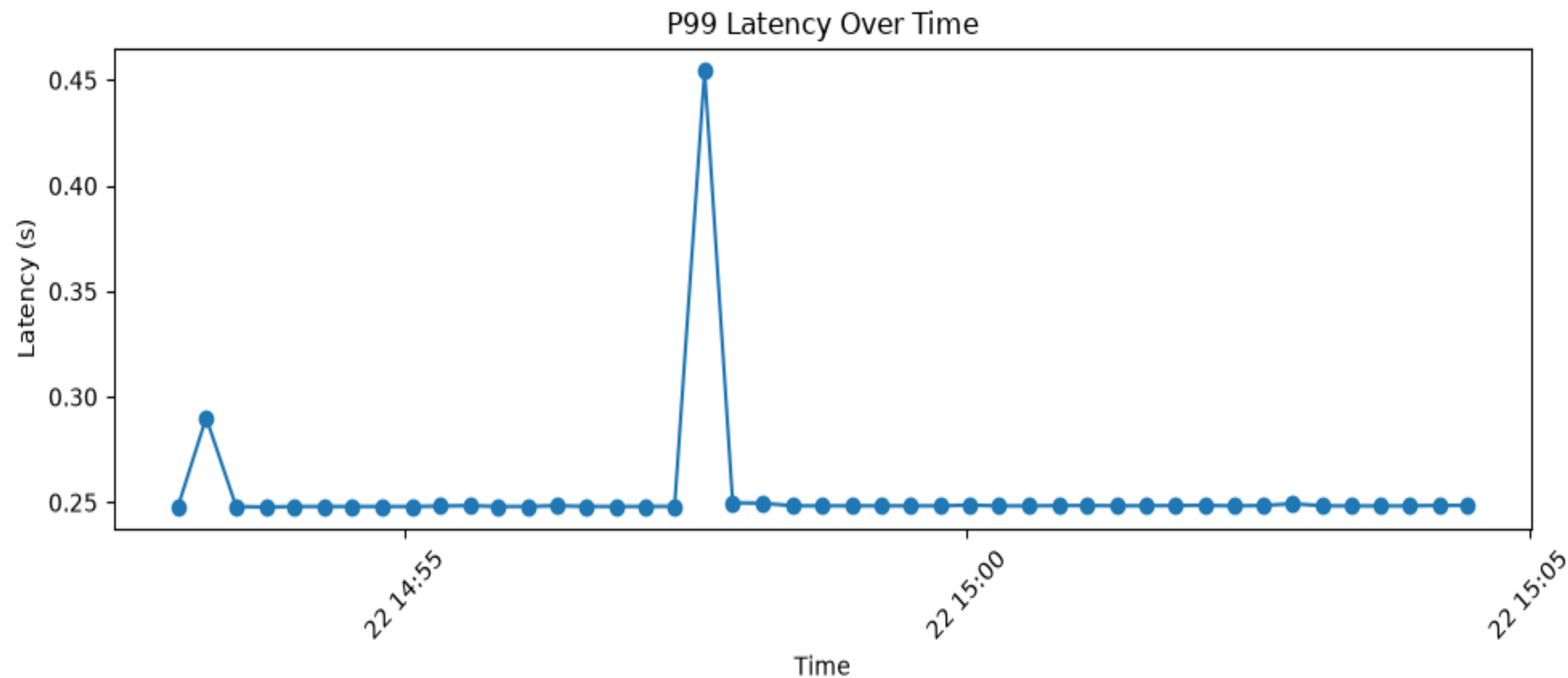
    if p99_latency is not None and p99_latency > 0.5:
        needed = max(current_replicas + 2, current_replicas + int(queue_size / 50))
        return min(needed, MAX_REPLICAS)

    if queue_size > 100:
        return min(current_replicas + 2, MAX_REPLICAS)

    if queue_size > 20:
        return min(current_replicas + 1, MAX_REPLICAS)

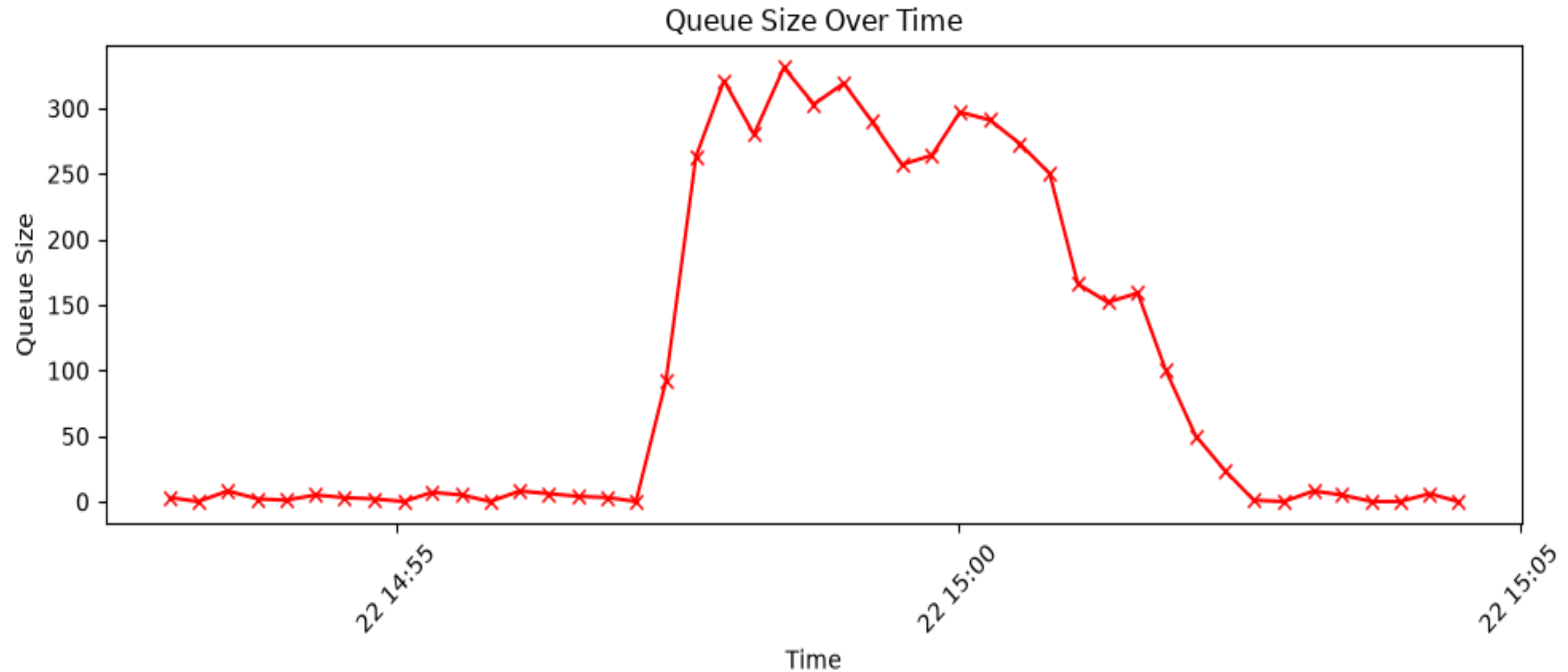
    return current_replicas
```

P99 latency with workload.txt



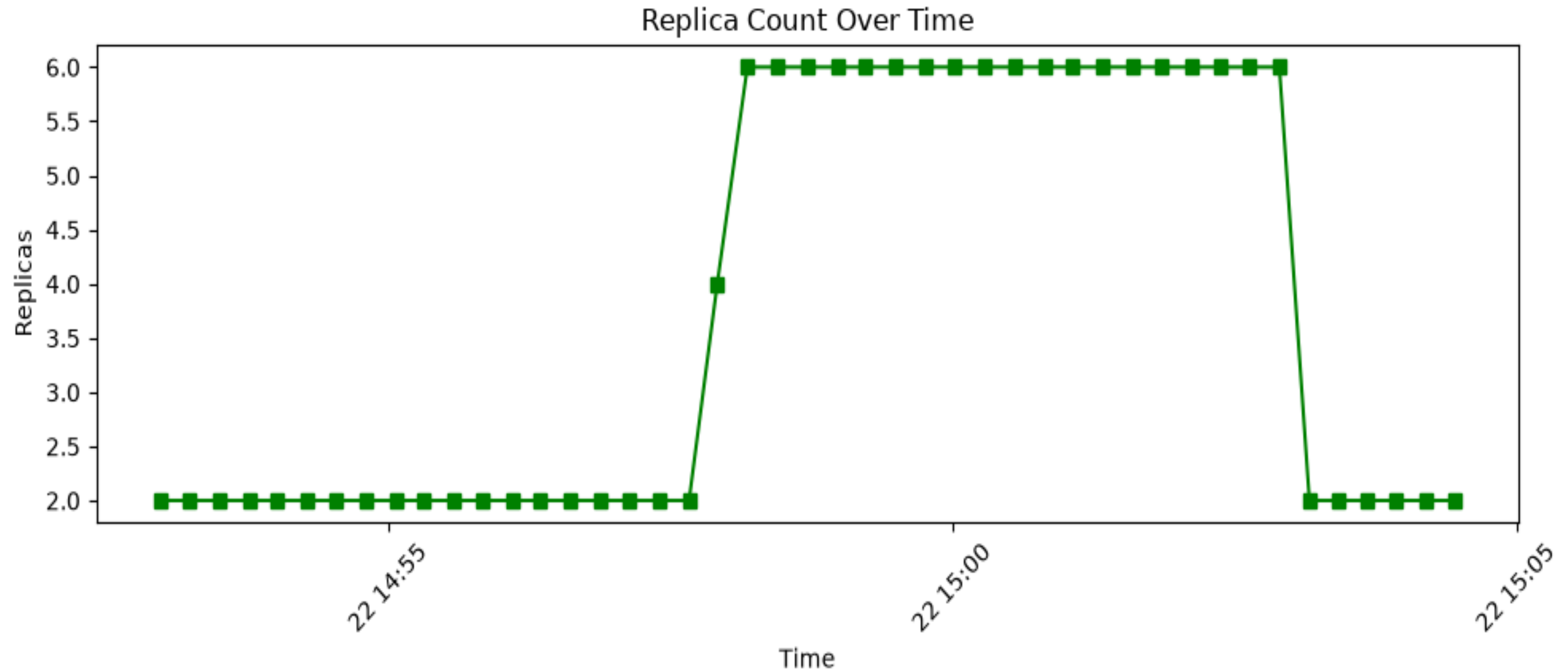
P99 latency stayed below 0.5 s

Queue size with workload.txt



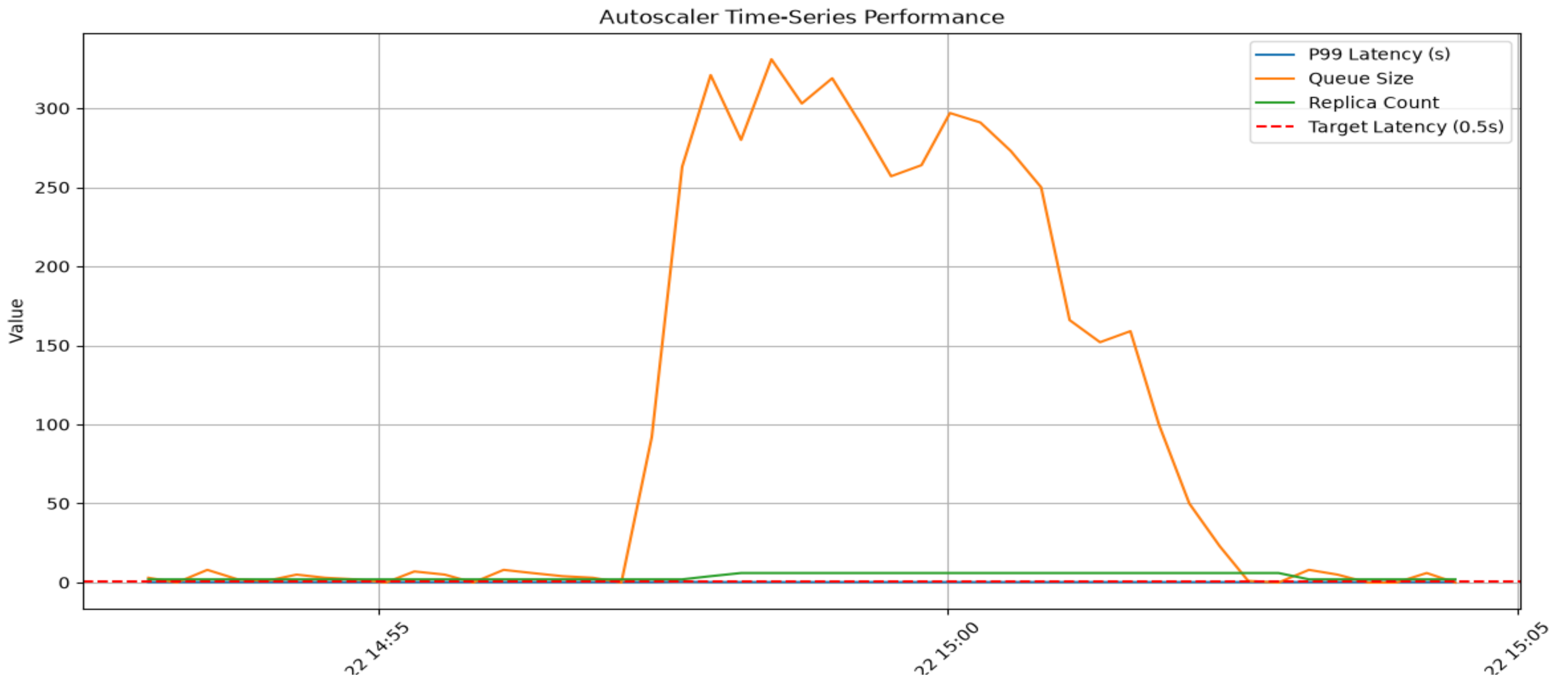
Queue size eventually returned to 0

Replica count with workload.txt



Replicas scaled up from 1 $\rightarrow$ 6 then scaled down from 6 $\rightarrow$ 1

Autoscaler Time-Series Performance of Custom HPA

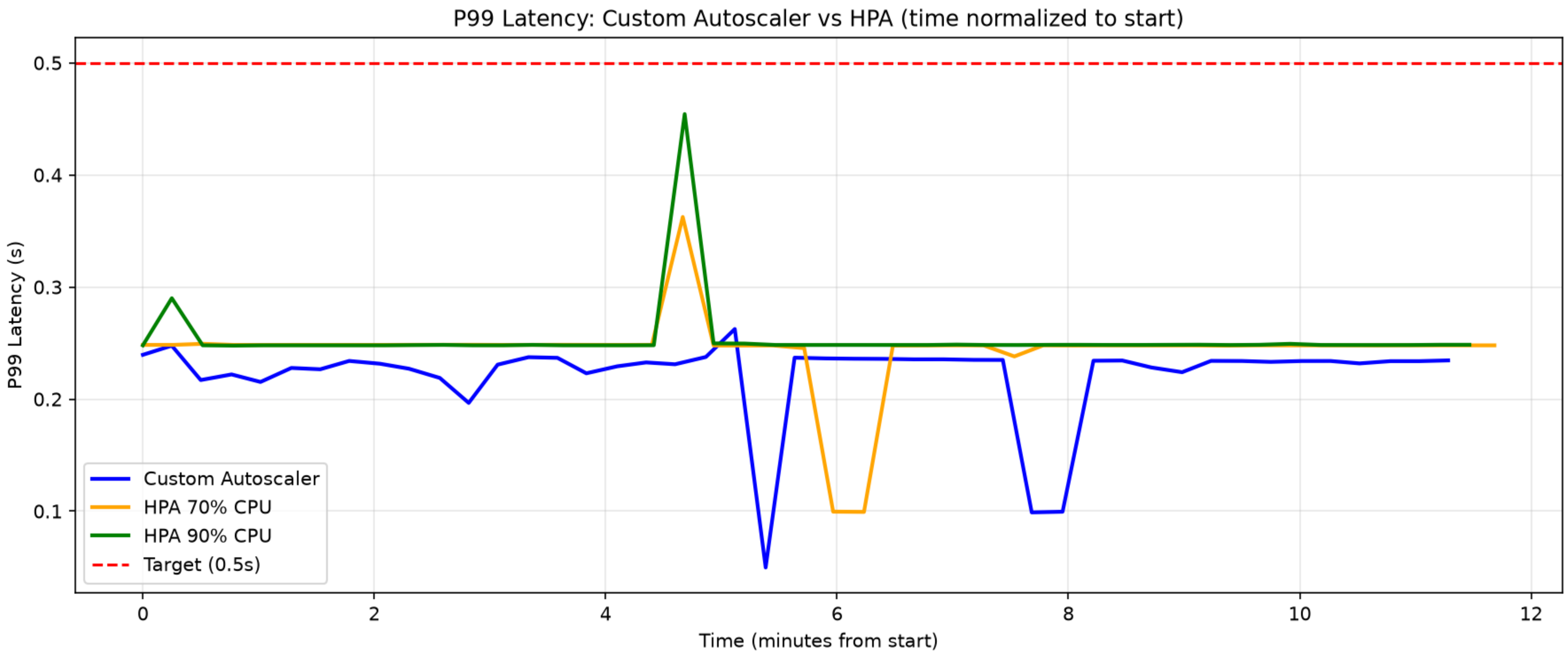


Logs with workload.txt

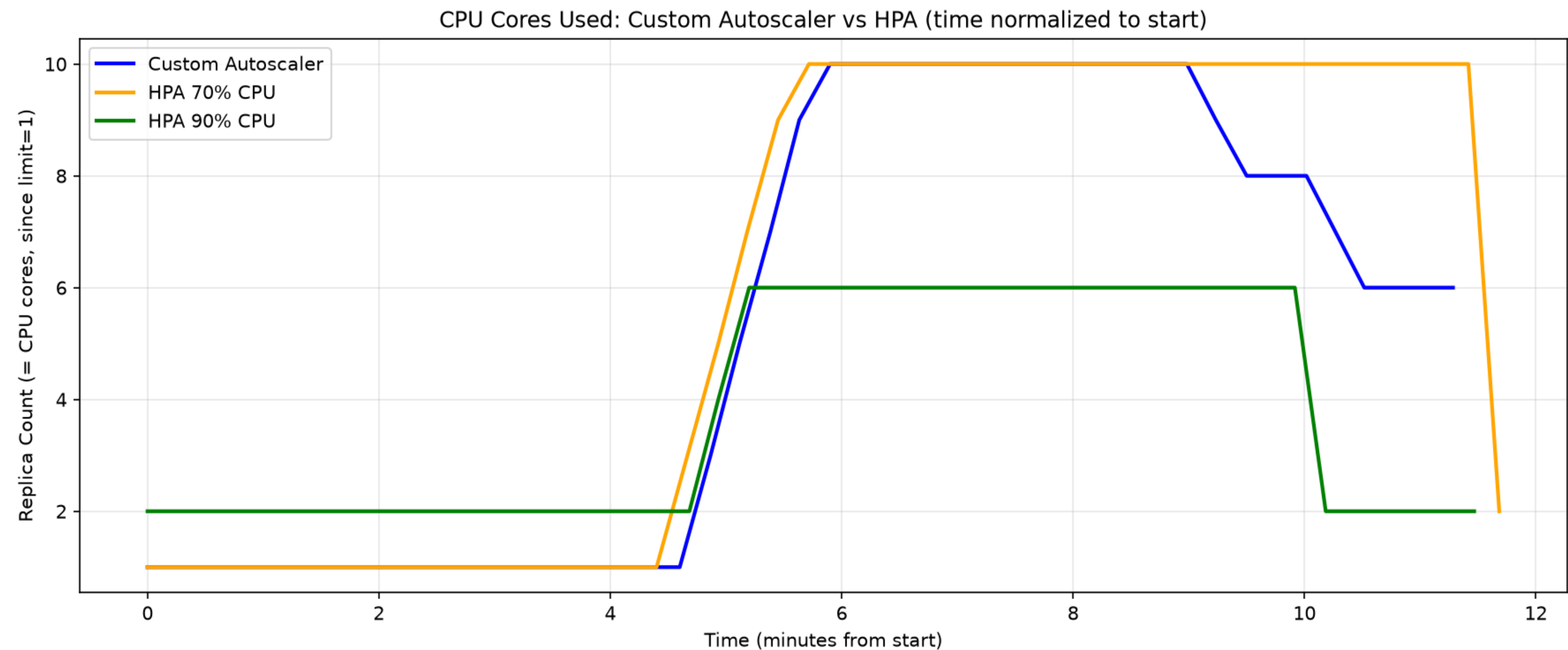
Timestamp	P99 Latency	Queue Size	Replicas	Action
2026-06-22T14:52:43	N/A	0	2	SCALE DOWN -> 1 replicas
2026-06-22T14:52:59	0.248s	3	2	No scaling needed
2026-06-22T14:53:14	0.290s	0	2	SCALE DOWN -> 1 replicas
2026-06-22T14:53:30	0.248s	8	2	No scaling needed
2026-06-22T14:53:46	0.248s	2	2	No scaling needed
2026-06-22T14:54:01	0.248s	1	2	No scaling needed
2026-06-22T14:54:17	0.248s	5	2	No scaling needed
2026-06-22T14:54:32	0.248s	3	2	No scaling needed
2026-06-22T14:54:48	0.248s	2	2	No scaling needed
2026-06-22T14:55:04	0.248s	0	2	SCALE DOWN -> 1 replicas
2026-06-22T14:55:19	0.248s	7	2	No scaling needed
2026-06-22T14:55:35	0.248s	5	2	No scaling needed
2026-06-22T14:55:50	0.248s	0	2	SCALE DOWN -> 1 replicas
2026-06-22T14:56:06	0.248s	8	2	No scaling needed
2026-06-22T14:56:21	0.248s	6	2	No scaling needed
2026-06-22T14:56:37	0.248s	4	2	No scaling needed
2026-06-22T14:56:53	0.248s	3	2	No scaling needed
2026-06-22T14:57:08	0.248s	0	2	SCALE DOWN -> 1 replicas
2026-06-22T14:57:24	0.248s	92	2	SCALE UP -> 3 replicas
2026-06-22T14:57:40	0.454s	263	2	SCALE UP -> 4 replicas
Timestamp	P99 Latency	Queue Size	Replicas	Action
2026-06-22T14:57:55	0.250s	321	4	SCALE UP -> 6 replicas
2026-06-22T14:58:11	0.250s	280	6	SCALE UP -> 8 replicas
2026-06-22T14:58:27	0.248s	331	6	SCALE UP -> 8 replicas
2026-06-22T14:58:43	0.248s	303	6	SCALE UP -> 8 replicas
2026-06-22T14:58:59	0.248s	319	6	SCALE UP -> 8 replicas
2026-06-22T14:59:14	0.248s	290	6	SCALE UP -> 8 replicas
2026-06-22T14:59:30	0.248s	257	6	SCALE UP -> 8 replicas
2026-06-22T14:59:46	0.248s	264	6	SCALE UP -> 8 replicas
2026-06-22T15:00:01	0.249s	297	6	SCALE UP -> 8 replicas
2026-06-22T15:00:17	0.248s	291	6	SCALE UP -> 8 replicas
2026-06-22T15:00:33	0.248s	273	6	SCALE UP -> 8 replicas
2026-06-22T15:00:49	0.248s	250	6	SCALE UP -> 8 replicas
2026-06-22T15:01:04	0.248s	166	6	SCALE UP -> 8 replicas
2026-06-22T15:01:20	0.248s	152	6	SCALE UP -> 8 replicas
2026-06-22T15:01:36	0.248s	159	6	SCALE UP -> 8 replicas
2026-06-22T15:01:51	0.248s	100	6	SCALE UP -> 7 replicas
2026-06-22T15:02:07	0.249s	50	6	SCALE UP -> 7 replicas
2026-06-22T15:02:23	0.248s	23	6	SCALE UP -> 7 replicas
2026-06-22T15:02:38	0.248s	1	6	No scaling needed
2026-06-22T15:02:54	0.249s	0	6	SCALE DOWN -> 5 replicas

Comparison of Custom HPA with HPA70 and HPA90

P99 Latency: Custom Autoscaler vs HPA70 vs HPA90

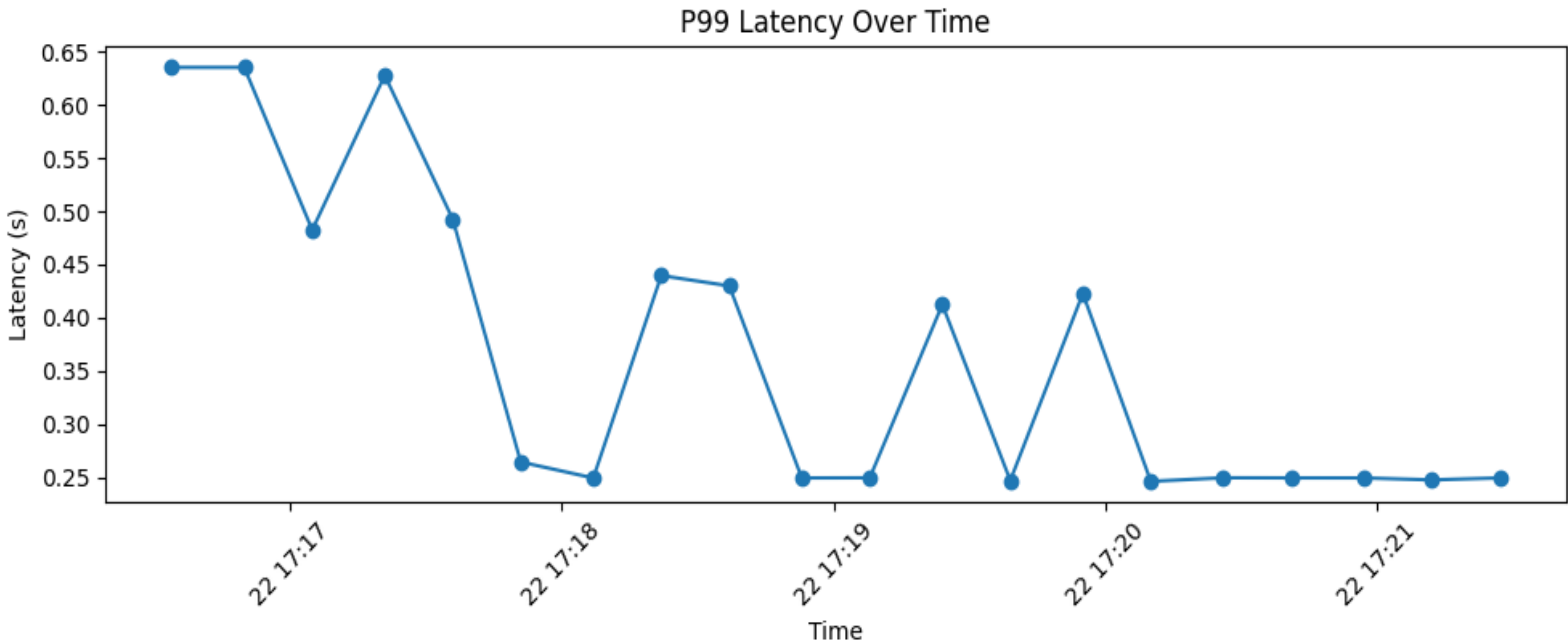


Replica count Custom Autoscaler vs HPA

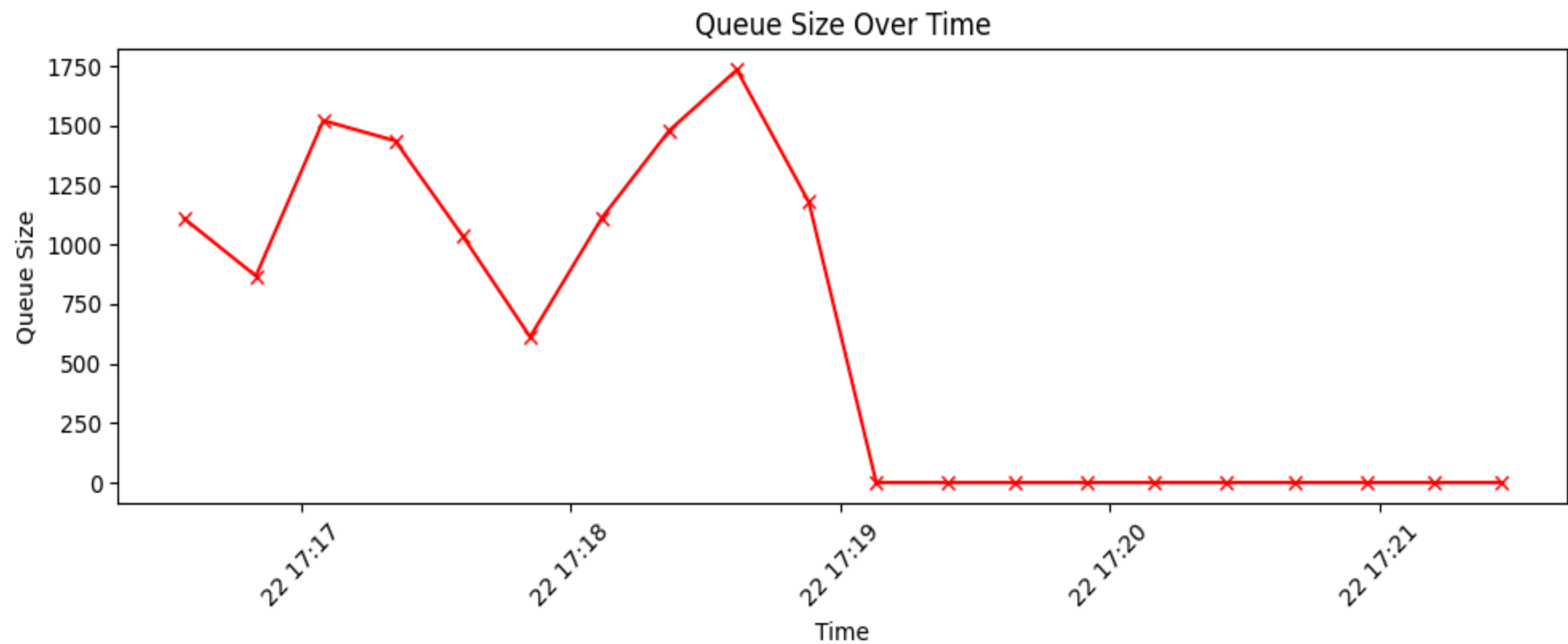


Additional experiments using heavy_load.txt

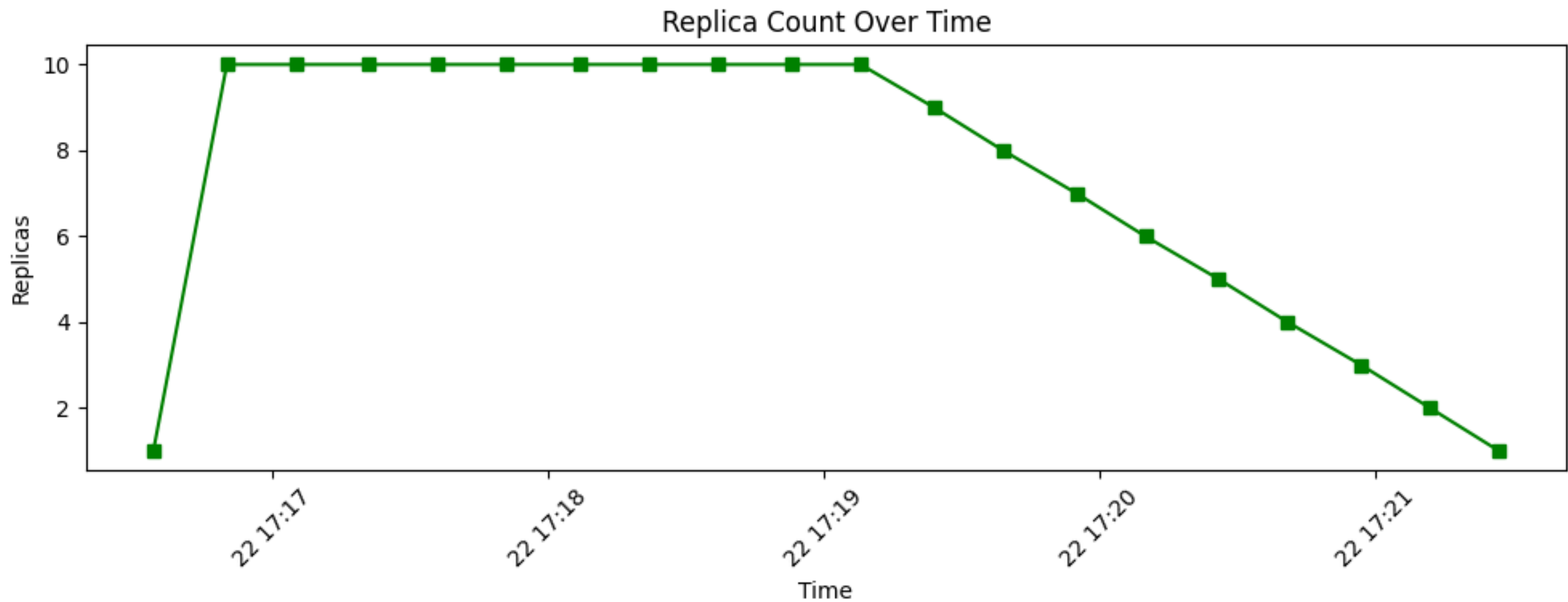
P99 latency with heavy_load.txt



Queue size with heavy_load.txt



Replica count with heavy_load.txt



Logs with heavy_load.txt

Timestamp	P99 Latency	Queue Size	Replicas	Action
2026-06-22T17:16:19	N/A	0	1	No scaling needed
2026-06-22T17:16:34	0.635s	1110	1	SCALE UP -> 10 replicas
2026-06-22T17:16:50	0.635s	868	10	No scaling needed
2026-06-22T17:17:05	0.482s	1521	10	No scaling needed
2026-06-22T17:17:21	0.627s	1436	10	No scaling needed
2026-06-22T17:17:36	0.493s	1036	10	No scaling needed
2026-06-22T17:17:51	0.265s	612	10	No scaling needed
2026-06-22T17:18:07	0.250s	1113	10	No scaling needed
2026-06-22T17:18:22	0.440s	1480	10	No scaling needed
2026-06-22T17:18:37	0.430s	1734	10	No scaling needed
2026-06-22T17:18:53	0.250s	1179	10	No scaling needed
2026-06-22T17:19:08	0.250s	0	10	SCALE DOWN -> 9 replicas
2026-06-22T17:19:24	0.412s	0	9	SCALE DOWN -> 8 replicas
2026-06-22T17:19:39	0.247s	0	8	SCALE DOWN -> 7 replicas
2026-06-22T17:19:55	0.423s	0	7	SCALE DOWN -> 6 replicas
2026-06-22T17:20:10	0.247s	0	6	SCALE DOWN -> 5 replicas
2026-06-22T17:20:26	0.250s	0	5	SCALE DOWN -> 4 replicas
2026-06-22T17:20:41	0.250s	0	4	SCALE DOWN -> 3 replicas
2026-06-22T17:20:57	0.250s	0	3	SCALE DOWN -> 2 replicas
2026-06-22T17:21:12	0.248s	0	2	SCALE DOWN -> 1 replicas
Timestamp	P99 Latency	Queue Size	Replicas	Action
2026-06-22T17:21:27	0.250s	0	1	No scaling needed
2026-06-22T17:21:43	N/A	0	1	No scaling needed

Danke